

Hospital Performance Analysis Report

Overview

This report presents an analysis of hospital performance based on patient distribution, workforce allocation, and financial metrics. The data focuses on the number of patients, doctors, nurses, appointments, and revenue, with additional insights into appointments, treatments, and revenue.

Key Metrics

- **Total Patients:** 50,000
- **Total Doctors:** 8,500
- **Total Nurses:** 17,000
- **Total Departments:** 16
- **Specialties:** 639
- **Total Appointments:** 80,000
- **Total Bills Processed:** 105,000
- **Total Medications Dispensed:** 158,000
- **Total Revenue:** 265 million

Patient Distribution Analysis

Age Group Distribution

Patients are distributed fairly evenly across age groups which highlights a fair participation of the various age group engagements, except for the 91-100 age group, which accounts for only 0.8%. The highest representation is from the 81-90 age group with 11.4% of the total patient population.

Implications:

- The 81-90 age group demands significant resources for geriatric care, including specialized services for age-related health issues.
- Low engagement of 91-100-year-olds might indicate barriers to access or reduced healthcare-seeking behaviour.

Recommendations:

- Expand geriatric care facilities and programs to cater to the 81-90 age group's increasing needs.
- Investigate possible barriers encountered by the 91-100 age group and introduce tailored outreach efforts, such as home-based services.

Gender Distribution

- **Male Patients:** 47.9%
- **Female Patients:** 48.3%
- **Non-Binary Patients:** 3.87%

Implications:

- This near-equal gender distribution suggests that healthcare needs are evenly spread across males and females reflecting inclusivity in healthcare services.

Recommendations:

- Maintain gender-equitable healthcare practices and resources for male and female patients.
- Invest in creating safe, welcoming environments for non-binary individuals by training staff on gender sensitivity and inclusivity.
- Collect feedback from non-binary patients to better understand their needs and barriers.

Workforce Distribution Analysis

Doctors by experience and specialties

There seems to be a tightly fair distribution of doctors across the various ages of experience, except in the 32-40 years which make up only 23.53% of the entire doctor's population and with highest concentration of doctors within the Incentivize E-Business E-Commerce specialty raking up to 6.78% of the entire doctor's population within the hospital.

Implications:

- A balanced distribution across experience levels suggests sufficient mentorship and talent development opportunities.
- Specialized departments with high doctor concentrations may indicate targeted areas of patient demand or potential resource strain.

Recommendations:

- Provide continuous training and leadership pathways to retain experienced doctors (31-40 years group).
- Monitor workload distribution in departments with high doctor concentrations to avoid burnout or inefficiencies.

Nurses distribution by shift patterns and specialties

- **By Shifts:** There appears to be a fair distribution among nurses on various shifts, with a slight decline in nurses on the morning shift, accounting for 32.77% as against 33.76% and 33.47% nurses on evening and night shift, respectively
- **By Specialties:** There also seems to be a wide distribution of the nurses' population across the vast number of specialties, with the highest number of nurses (1,102) within the Seize Proactive Deliverables specialty, which only account for 6.48% of the entire nurses' population.

Implications:

- Slightly lower staffing during morning shifts could pose challenges during peak patient interaction hours.
- Even distribution across departments showcases effective staffing but may not reflect individual department needs.

Recommendations

- Redistribute nurse staffing to ensure equal coverage during peak morning hours.

- Conduct periodic reviews of department-specific workloads to adjust staffing dynamically.

Appointment Analysis

Age group: There was an even distribution of appointment across various age -groups analysed with a slight increase within the 81 – 90 age group range which represents 11.67% of the entire patient appointment population and a drastic decline within the age group 91 – 100 which represents 0.87% of the entire patient appointment population.

Gender: there was an almost equal distribution of the patient appointment population amongst the female (48.26%) and male (47.94%) genders, which was in contrast to 3.8% seen within the non-binary gender population.

Month: January recorded the highest peak of the appointments with 9.67%, and the lowest in February with 7.82%. There was a seemingly fair distribution within the other months of the year. This analysis also mirrored the treatment analysis, and can be said that most of the patients received treatments on appointments.

Years: There has also been a tightly even distribution of appointment over the years with its highest at 2022 and 2021 with 16.67% and 16.66% respectively and a sharp decline in 2025 with only 1.4%.

Status: 69.2% of the appointment schedules have been completed with 20.22% scheduled and 10.16% cancelled.

Implications:

- Discrepancies with age groups 91 – 100 and non-Binary accessing appointments, and possibly a reflection on treatment.
- Increased workload in January on the staff workforce against other months of the year.
- 10.16% patient population cancelled appointments.

Recommendations:

- Collect feedback from non-binary patients and the minority age group to better understand their needs and barriers.
- Support workforce development through balanced resource allocation and staffing adjustments in months of high engagement.
- Conduct surveys, investigate, and obtain feedback as to the events and possible causes of appointment cancellations.
- Analyse patient demographics in years with fewer appointments to identify potential barriers.
- Introduce sending appointment reminders for pending appointments.

Payment and Revenue Analysis**Payment Breakdown**

- Paid (60%): Represents strong revenue recovery, supporting operational needs.
- Pending (20%) and Unpaid (20%): Combined 40% indicates recurring challenges in closing payment gaps.

Implications:

- Pending payments represent future revenue opportunities but require active follow-ups.
- Unpaid payments risk financial instability, highlighting the need for targeted collection strategies.

Recommendations:

- Optimize operational strategies to enhance patient engagement and payment rates.
- Implement automated reminders for patients with pending payments.
- Offer options like installment plans for unpaid bills.
- Simplify billing procedures and introduce alternative payment portals.

Revenue Trends

- Revenue distribution remains tightly consistent across years (16.28% –16.50%), suggesting steady financial health.
- Slightly lower revenue in 2022 (16.36%), despite the highest appointments, highlights inefficiencies in converting appointments into payments.

Implications:

- Financial stability across years is promising but lacks significant growth. Increasing paid revenues from high-appointment years could improve profitability.

Recommendations:

- Deploy seamless payment systems to ensure appointments translate directly into revenues.
- Incentivize early payments for patients completing appointments in bulk.
- Conduct thorough audits in years like 2022, where revenue dips occurred despite operational successes.
- Analyse and replicate trends from high-performing years.

Conclusion

The hospital demonstrates strong performance across various metrics, with a well-distributed workforce and patient demographics. Necessary adjustment at peak time will help facilitate the workflow. The hospital should deploy means to recovery unpaid revenues as this will inversely boost the growth of healthcare services and efficiency.